William Pearman

Research Fellow

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Auckland, New Zealand

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About me

Molecular and microbial ecologist working on population genomics and the evolution of host-microbe (holobiont) relationships.

Scholar Stats

477 citations | h-index: 8 | i10-index: 8

I am a broadly trained molecular and microbial ecologist specialising in population genomics and microbial ecology, developing eco-evolutionary simulations of host-microbe (holobiont) relationships tested against observational and experimental data.

Education

2023 — PhD Marine Science — University of Otago
Dunedin, New Zealand

2020 — MNatSci (Distinction) Genetics — Massey University
Auckland, New Zealand

2018 — BNatSci Genetics — Massey University
Auckland, New Zealand

Positions

2024– — Research Fellow — University of Auckland
Auckland, New Zealand

2023 — Assistant Research Fellow / Lab Manager — University of Otago
Dunedin, New Zealand

Distinctions, Grants & Memberships

- 2026 — Visiting Scholar - KU Leuven
- 2025-present — Co-chair - University of Auckland Faculty of Science Research Fellow Society
- 2025-present — Member - University of Auckland Faculty of Science Research Committee
- 2025 — Mana Tūāpapa Future Leader Fellowship ($820,000 NZD) - Independent 4-year fellowship to explore Daphnia microbial ecology
- 2025 — Te Pūnaha Matatini Seed Fund ($4,880 NZD)
- 2025 — ISME Scholar Mobility Award ($3,000 NZD)

Selected Publications

1. Pearman, W.S., Freed, N.E. & Silander, O.K. (2020). Testing the advantages and disadvantages of short- and long-read eukaryotic metagenomics using simulated reads. *BMC Bioinformatics*, 21(1), 1-15. DOI

7. Pearman, W.S., Urban, L. & Alexander, A. (2022). Commonly used Hardy-Weinberg equilibrium filtering schemes impact population structure inferences using RADseq data. *Molecular Ecology Resources*, 22(7), 2599-2613. DOI

5. Fraser, C.I., Dutoit, L., Morrison, A.K., Pardo, L.M., Smith, S.D.A., Pearman, W.S., Parvizi, E., Waters, J. & Macaya, E.C. (2022). Southern Hemisphere coasts are biologically connected by frequent, long-distance rafting events. *Current Biology*, 32(14), 3154-3160.e3. DOI

2. Arranz, V., Pearman, W.S., Aguirre, J.D. & Liggins, L. (2020). MARES, a replicable pipeline and curated reference database for marine eukaryote metabarcoding. *Scientific Data*, 7(1), 209. DOI

9. Pearman, W.S., Duffy, G.A., Liu, X.P., Gemmell, N.J., Morales, S.E. & Fraser, C.I. (2024). Macroalgal microbiome biogeography is shaped by environmental drivers rather than geographical distance. *Annals of Botany*, 133(1), 169-182. DOI

6. Liu, X.P., Duffy, G.A., Pearman, W.S., Pertierra, L.R. & Fraser, C.I. (2022). Meta-analysis of Antarctic phylogeography reveals strong sampling bias and critical knowledge gaps. *Ecography*, 2022(12), e06312. DOI

13. Pearman, W.S., Arranz, V., Carvajal, J.I., Whibley, A., Liau, Y., Johnson, K., Gray, R., Treece, J.M., Gemmell, N.J., Liggins, L., Fraser, C.I., Jensen, E.L. & Green, N.J. (2024). A cry for kelp: Evidence for polyphenolic inhibition of Oxford Nanopore sequencing of brown algae. *Journal of Phycology*, 60(6), 1601-1610. DOI

3. Pearman, W.S., Wells, S.J., Silander, O.K., Freed, N.E. & Dale, J. (2020). Concordant geographic and genetic structure revealed by genotyping-by-sequencing in a New Zealand marine isopod. *Ecology and Evolution*, 10(24), 13624-13639. DOI